

Cairo University

Faculty of Computing and Artificial Intelligence – General Program



Project Title : The Renewable

Energy Grid & Maintenance Network

Project ID : 33



PHASE 3

TEAM INFO

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Requirements

A. Business Rules & Logic

- An Energy Site houses multiple Power Units (1:M).
- A Field Technician can hold multiple Certifications, and a Certification can belong to multiple Technicians (M:N).
- A Technician conducts Inspection Rounds at specific Energy Sites.
- During an Inspection Round, multiple Unit Readings are generated to track the efficiency and status of the Power Units.
- When maintenance is required, Components (spare parts) are replaced within specific Power Units, tracked by their Serial Number and Replacement Date.

B. System Functionalities

- Managing energy sites and the power units installed within them.
- Registering field technicians and their specific technical certifications.
- Scheduling and logging periodic inspection rounds for each site.
- Tracking component replacements and spare part inventory.
- Monitoring unit efficiency readings and identifying underperforming units.

C. System Inquiries

1. Which power unit "manufacturer" had the highest number of efficiency readings below average?
2. Which energy site had no inspection rounds conducted during the last month?
3. Who was the technician who completed the maximum number of inspections last month?
4. Identify power units that did not require any component replacements last month.
5. What were the specific components installed at each energy site last month?
6. For each technician, retrieve their profile and the total number of power units they inspected.

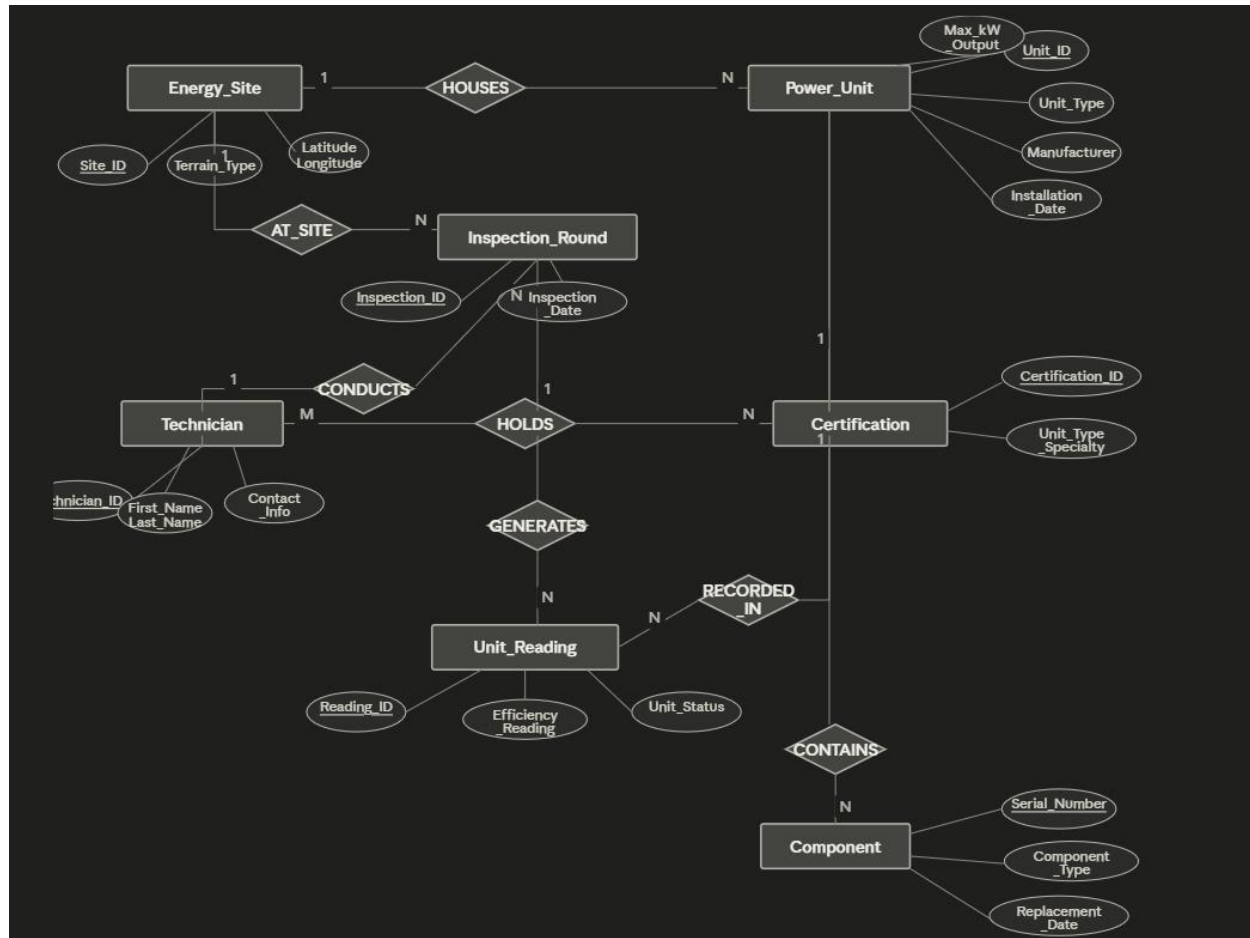
Github

Link :

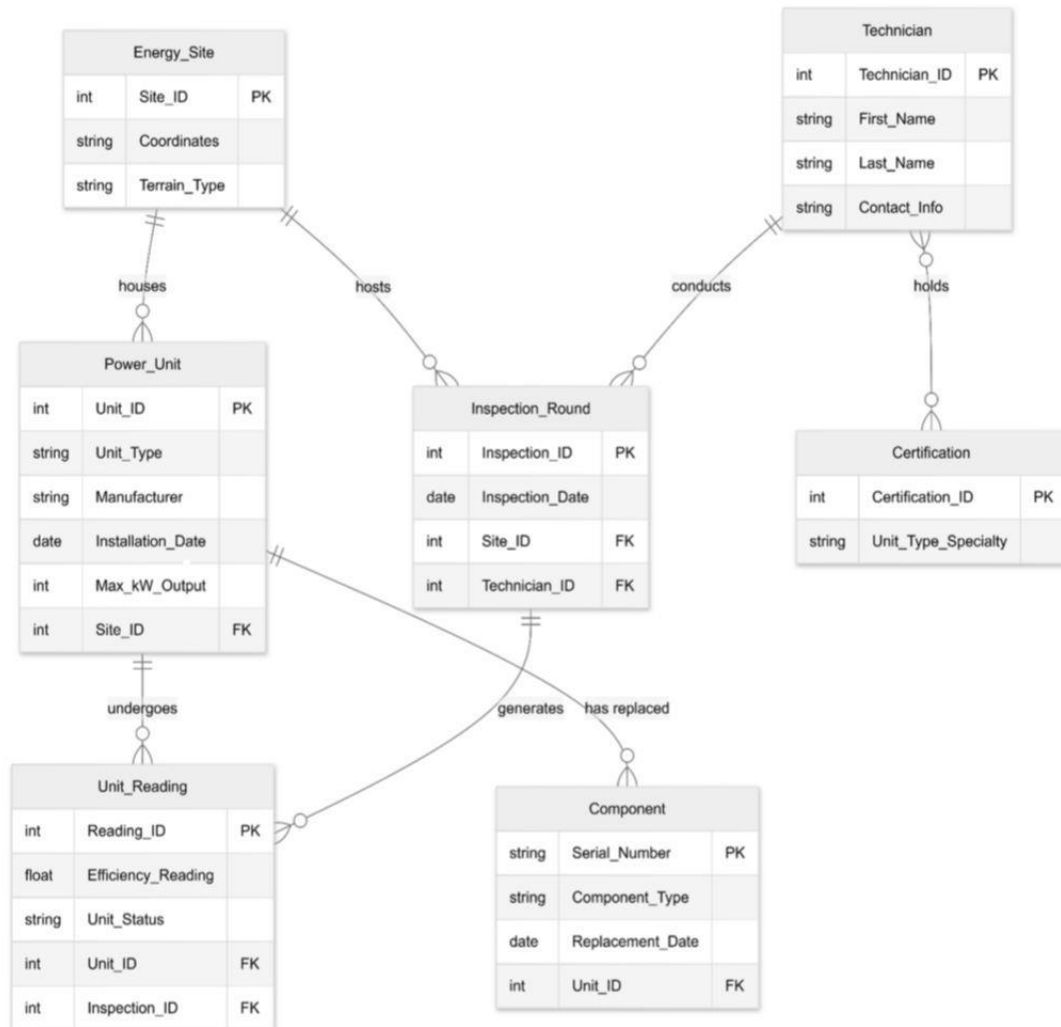
<https://github.com/Mohaitham1/renewable-energy-grid-db>

Used To Store all Our Project files and assets , and version tracking between Co Team members

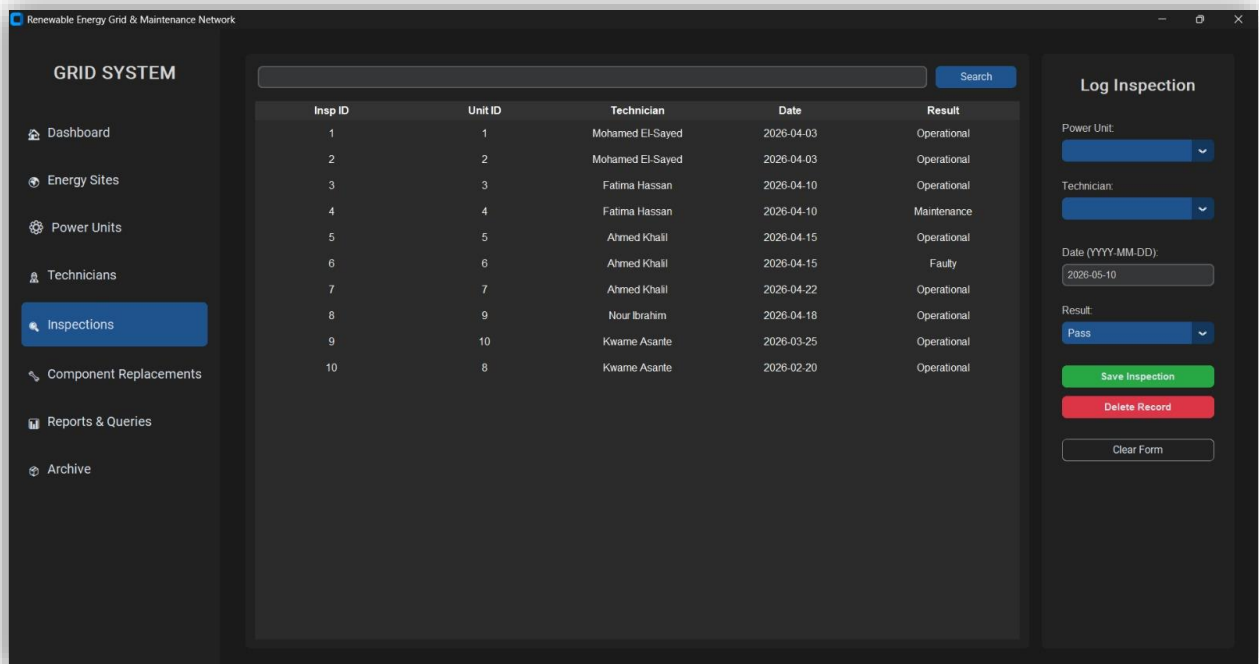
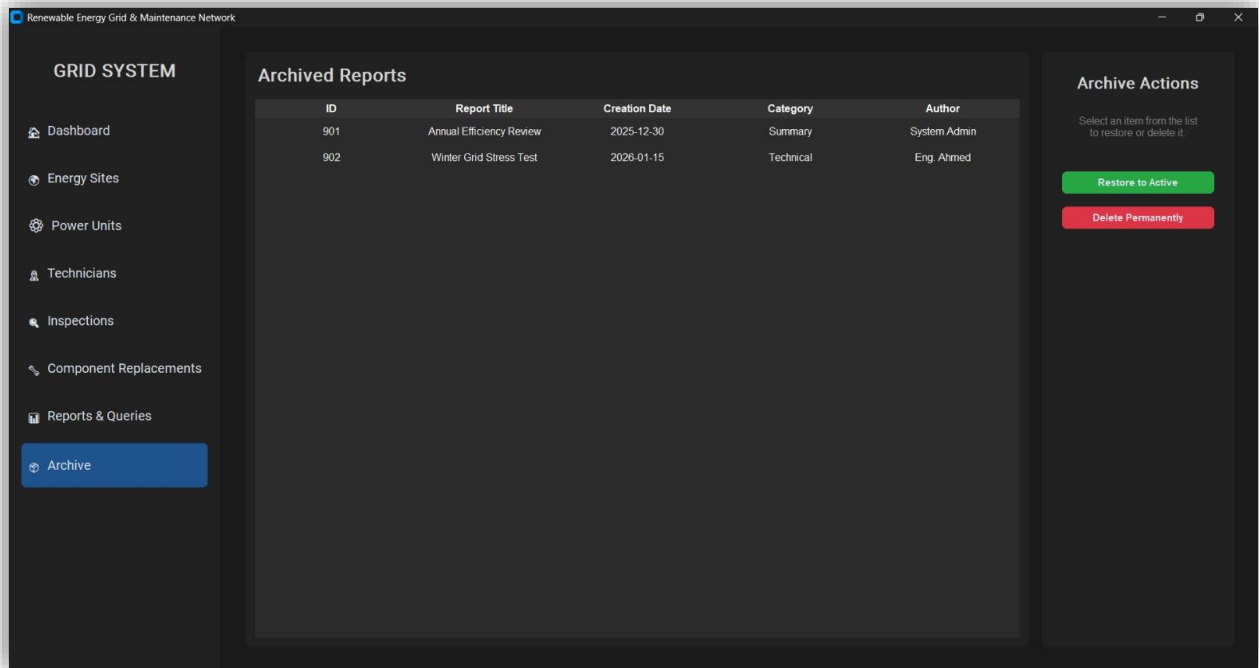
Conceptual ERD



Physical ERD



Program Screenshot



Renewable Energy Grid & Maintenance Network

GRID SYSTEM

Dashboard

Energy Sites

Power Units

Technicians

Inspections

Component Replacements

Reports & Queries

Archive

Search

Unit ID	Parent Site	Unit Type	Status	Max kW Output
1	Aswan Solar Complex	Solar	Operational	370.00
2	Aswan Solar Complex	Solar	Operational	370.00
3	Benban Solar Park	Solar	Operational	420.00
4	Benban Solar Park	Solar	Maintenance	420.00
5	Gulf of Suez Wind Farm	Wind	Operational	4500.00
6	Gulf of Suez Wind Farm	Wind	Faulty	4500.00
7	Hurghada Wind Station	Wind	Operational	3600.00
8	Gabal El-Zayt Wind Park	Wind	Operational	12000.00
9	Lake Nasser Solar Hub	Solar	Operational	390.00
10	Nile Delta Wind Farm	Wind	Operational	6000.00

Unit Details

Parent Site:

Unit Type:

PV Panel

Manufacturer:

Model (optional):

Installation Date (YYYY-MM-DD):

2020-05-12

Max kW Output:

0

Status:

Operational

Serial Number (optional):

Save Unit

Delete Selected

Renewable Energy Grid & Maintenance Network

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Search

ID	Site Name	Country	Region	Terrain	Established
1	Aswan Solar Complex	Egypt	Upper Egypt	Desert	2017-04-10
2	Benban Solar Park	Egypt	Aswan Governorate	Desert	2019-01-15
3	Gulf of Suez Wind Farm	Egypt	Suez Governorate	Coastal	2016-06-20
4	Hurghada Wind Station	Egypt	Red Sea	Coastal	2018-03-05
5	Gabal El-Zayt Wind Park	Egypt	Red Sea	Coastal	2015-11-30
6	Lake Nasser Solar Hub	Egypt	Upper Egypt	Desert	2020-07-01
7	Nile Delta Wind Farm	Egypt	Nile Delta	Flatland	2021-02-14
8	Nairobi Green Energy Park	Kenya	Nairobi County	Highland	2018-09-10
9	Cape Wind & Solar Station	South Africa	Western Cape	Coastal	2017-05-22
10	Sahel Wind Corridor	Niger	Tillabeni	Plains	2020-11-03

Site Details

Site Name:

Country:

Region (optional):

Terrain Type:

Desert

Latitude:

0.0

Longitude:

0.0

Established Date (YYYY-MM-DD, optional):

Add Site

Delete Selected

Clear Form

Renewable Energy Grid & Maintenance Network

GRID SYSTEM

- Dashboard
- Energy Sites
- Power Units
- Technicians
- Inspections
- Component Replacements**
- Reports & Queries
- Archive

ID	Part Name	Part Number	Site	Unit Type	Replacement Date
2	Inverter Module	INV-002	Benban Solar Park	Solar	2026-04-11
3	Gearbox Assembly	GBX-004	Gulf of Suez Wind Farm	Wind	2026-04-16
4	Wind Turbine Blade	WTB-003	Gulf of Suez Wind Farm	Wind	2026-04-16
5	Solar Panel Glass	SPG-001	Lake Nasser Solar Hub	Solar	2026-04-19
6	Junction Box	JBX-006	Lake Nasser Solar Hub	Solar	2026-04-19
7	Yaw Drive Motor	YDM-007	Gabal El-Zayt Wind Park	Wind	2026-02-21
8	Anemometer Sensor	ANM-005	Gabal El-Zayt Wind Park	Wind	2026-02-21
9	Solar Cell Array	SCA-008	Nile Delta Wind Farm	Wind	2026-01-15
10	Pitch Control System	PCS-009	Hurghada Wind Station	Wind	2026-03-09

Component Replacement

Spare Part:

Site:

Part Serial Number:

Replacement Date (YYYY-MM-DD):

Quantity Used:

Notes (optional):

Save Replacement

Delete Selected

Clear

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Tech ID	Full Name	Email	Phone	Hire Date	Status
1	Mohamed El-Sayed	mohamed.elsayed@renew-eg.com	+20-100-1001	2015-03-10	Active
2	Fatima Hassan	fatima.hassan@renew-eg.com	+20-111-1002	2016-07-20	Active
3	Ahmed Khalil	ahmed.khalil@renew-eg.com	+20-122-1003	2017-01-15	Active
4	Nour Ibrahim	nour.ibrahim@renew-eg.com	+20-100-1004	2018-05-01	Active
5	Youssef Mansour	youssef.mansour@renew-eg.com	+20-115-1005	2019-09-22	Active
6	Amira Abdallah	amira.abdallah@renew-eg.com	+20-128-1006	2020-02-14	Active
7	Kwame Asante	kwame.asante@renew-af.com	+233-24-2001	2018-11-05	Active
8	Aisha Diallo	aisha.diallo@renew-af.com	+221-77-2002	2019-04-30	Active
9	Tariq Osman	tariq.osman@renew-af.com	+249-91-2003	2017-08-18	Active
10	Lena Mwangi	lena.mwangi@renew-eg.com	+254-72-2004	2021-01-10	Inactive

Staff Manager

Full Name:

Email:

Phone:

Hire Date (YYYY-MM-DD):

Employment Status:

Save Technician

Delete Selected

Clear

